

Experiment 31: FIFO Page Replacement Algorithm

C Program

```
#include <stdio.h>

int main()
{
    int pages[50], frame[10];
    int n, f, i, j, k=0, faults=0, flag;

    printf("Enter number of pages: ");
    scanf("%d", &n);

    printf("Enter page reference string:\n");
    for(i=0; i<n; i++)
        scanf("%d", &pages[i]);

    printf("Enter number of frames: ");
    scanf("%d", &f);

    for(i=0; i<f; i++)
        frame[i] = -1;

    printf("\nPage\tFrames\n");

    for(i=0; i<n; i++)
    {
        flag=0;
        for(j=0; j<f; j++)
            if(frame[j]==pages[i]) { flag=1; break; }

        if(flag==0)
        {
            frame[k]=pages[i];
            k=(k+1)%f;
            faults++;
        }

        printf("%d\t", pages[i]);
        for(j=0; j<f; j++)
            if(frame[j]!=-1) printf("%d ", frame[j]);
            else printf("- ");
        printf("\n");
    }

    printf("\nTotal Page Faults = %d\n", faults);
    return 0;
}
```

Input / Output / Result

Input

Enter number of pages: 12
Enter page reference string:
1 2 3 4 1 2 5 1 2 3 4 5
Enter number of frames: 3

Output

Total Page Faults = 9

Result

FIFO Page Replacement Algorithm executed successfully.